

Machine Learning-Based Prediction of Diabetes

Using Clinical and Demographic Risk Factors

Irene Li • Yunelle Teng • Ziyi Bao

March 2026

Introduction



Problem Motivation

- Diabetes is a major global health issue
- Early identification is important for prevention
- Traditional screening methods rely on simple thresholds (an HbA1c level above 6.5 percent)
- Machine learning can capture complex nonlinear relationships

Project Objective

- Build and evaluate machine learning models to predict whether an individual has diabetes using demographic, lifestyle, and clinical features.
- Target Variable:
Diabetes (1 = Yes, 0 = No)

Dataset Overview

100,000

Patient Records

16

Features

0

Missing Values

Feature Categories

Demographic

Age, Gender, Location, Race

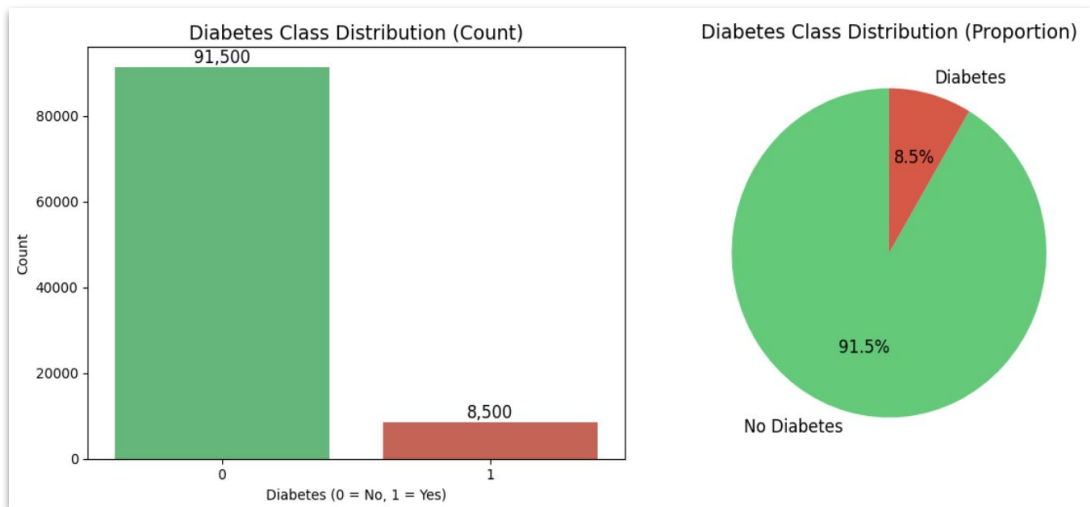
Clinical

BMI, HbA1c, Blood Glucose

Medical History

Hypertension, Heart Disease,
Smoking

Exploratory Data Analysis — Target Distribution

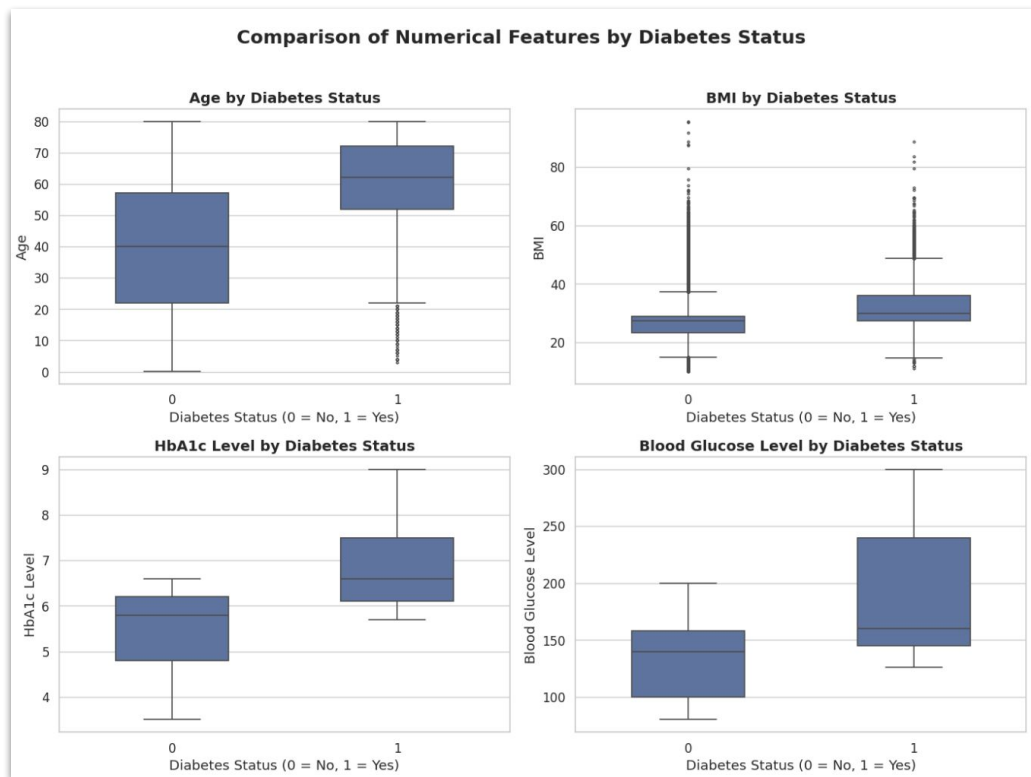


- Strong class imbalance
- A naive classifier predicting only the majority class would achieve 91.5% accuracy.

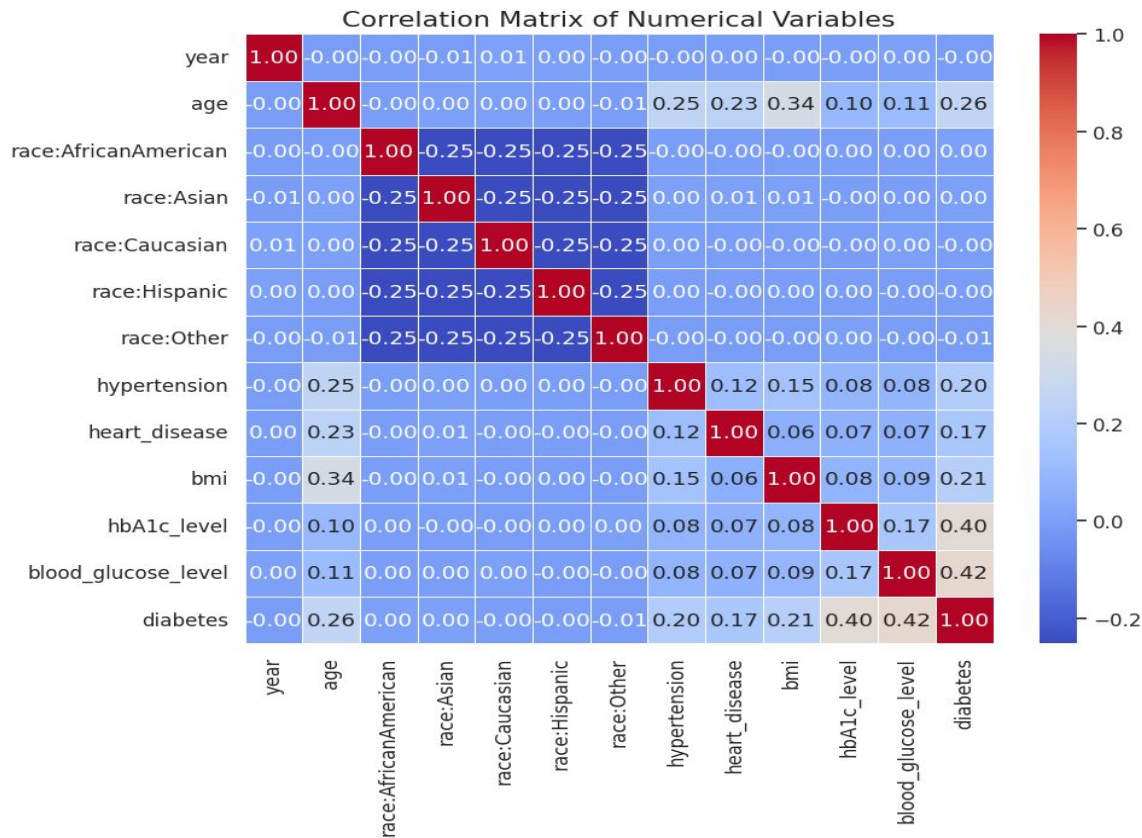
Exploratory Data Analysis — Numerical Features

Key patterns observed:

- **Age:** Diabetic patients tend to be older
- **BMI:** Higher BMI is associated with increased diabetes risk
- **HbA1c Level:** Strong separation between diabetic and non-diabetic patients
- **Blood Glucose Level:** Significantly higher among diabetic patients

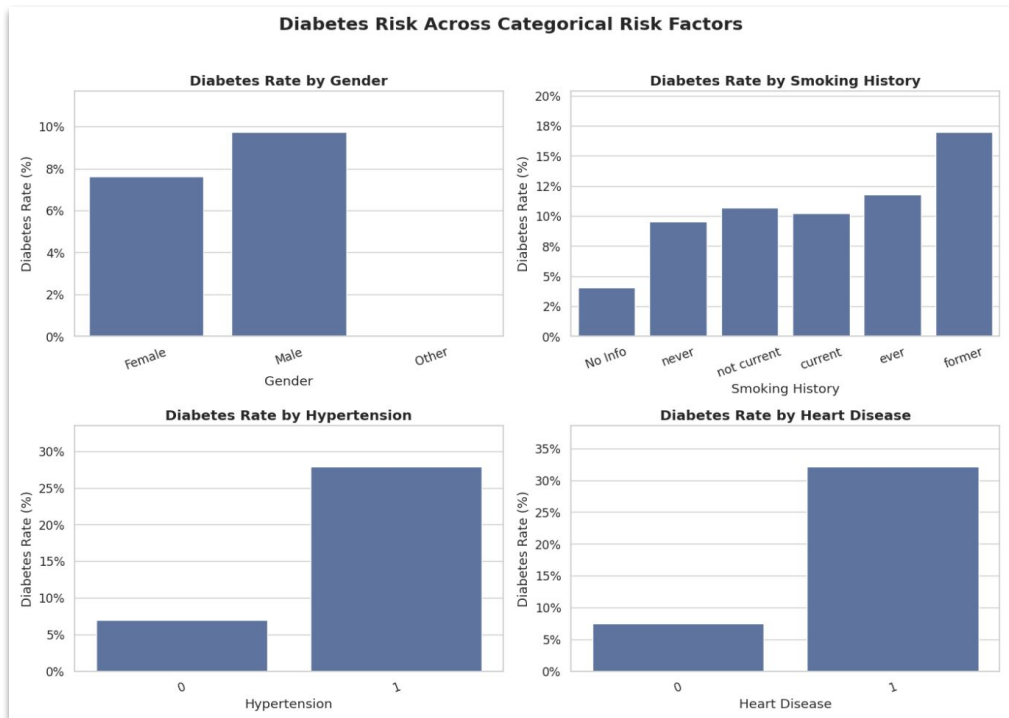


Exploratory Data Analysis — Correlation



- Year and race has no correlation with Diabetes.

Exploratory Data Analysis — Categorical Risk Factors



Key observations:

- **Gender:** Slightly higher diabetes prevalence among males
- **Smoking History:** Former smokers show higher diabetes rates
- **Hypertension:** People with hypertension have higher rate of diabetes
- **Heart Disease:** People with heart disease have significantly higher rate of diabetes

Feature Engineering

Feature Engineering Pipeline

16 columns → 14 columns | 100,000 → 99,982 rows

Dropped 7 Columns

year (corr: -0.003)
location (55 values, std=0.0075)
5 race columns (corr: ~0.00)

Encoded 2 Columns

gender → Binary (F=0, M=1)
smoking_history → Ordinal (0-5)
Dropped 18 "Other" gender rows

Transformed 1 Column

bmi → Capped at 99th percentile
Reduced outlier influence (max: 95.69)

Created 5 New Columns

Age bins: 0-18, 19-35, 36-50, 51-65, 66-80
One-hot encoded from continuous age

Final Dataset

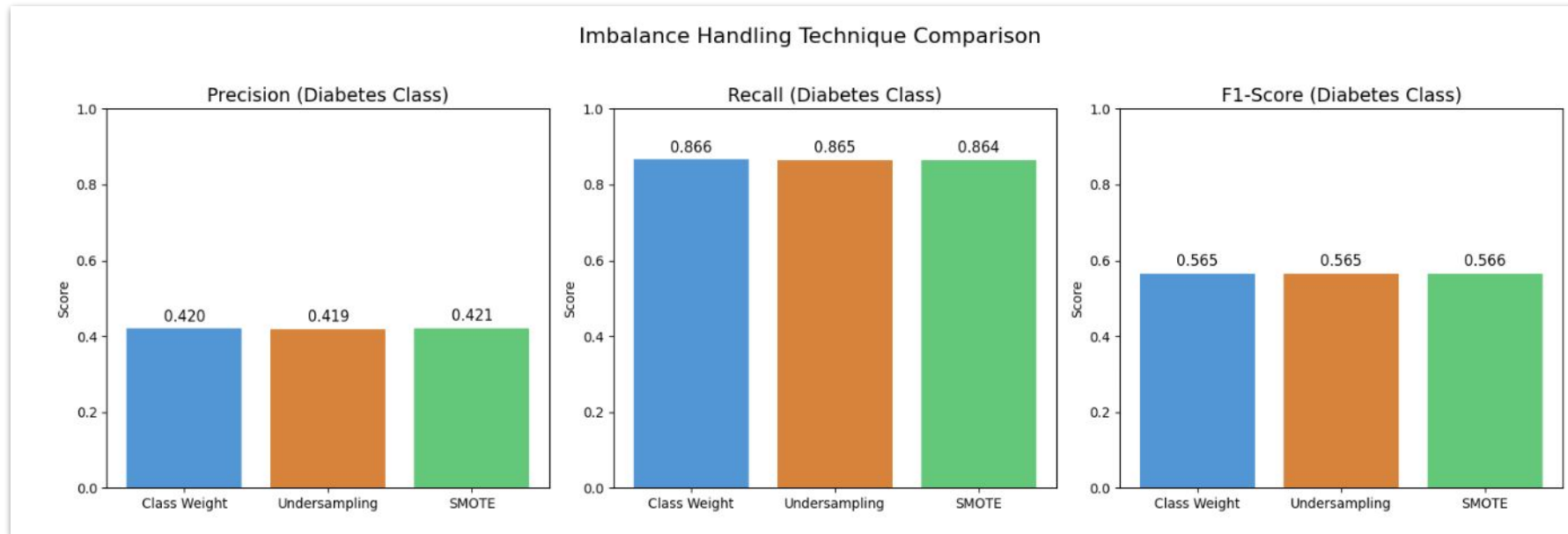
14 columns | 99,982 rows | All numeric | Model-ready
Target: diabetes (91.5% No / 8.5% Yes)

Methodology

- 80/20 stratified split
- Class Weighting to rebalance class
- 4 models (Logistic Regression, Decision Tree, Random Forest, XGBoost)
- 5-Fold Cross-Validation
- Prioritize recall; access precision, F1-score, and PR-Curve
- Threshold Tuning

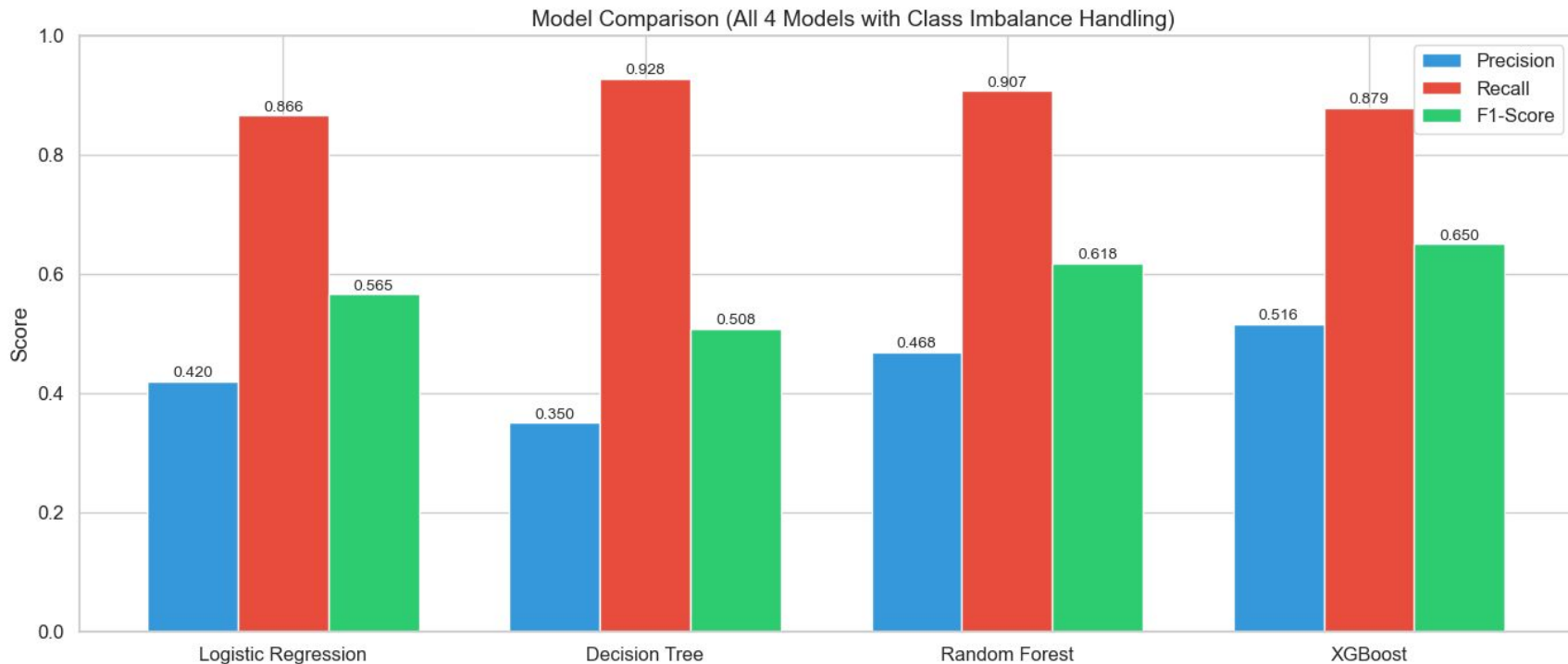
Handling Class Imbalance

We built a Logistic Regression to test which class rebalance technique works better on our data.

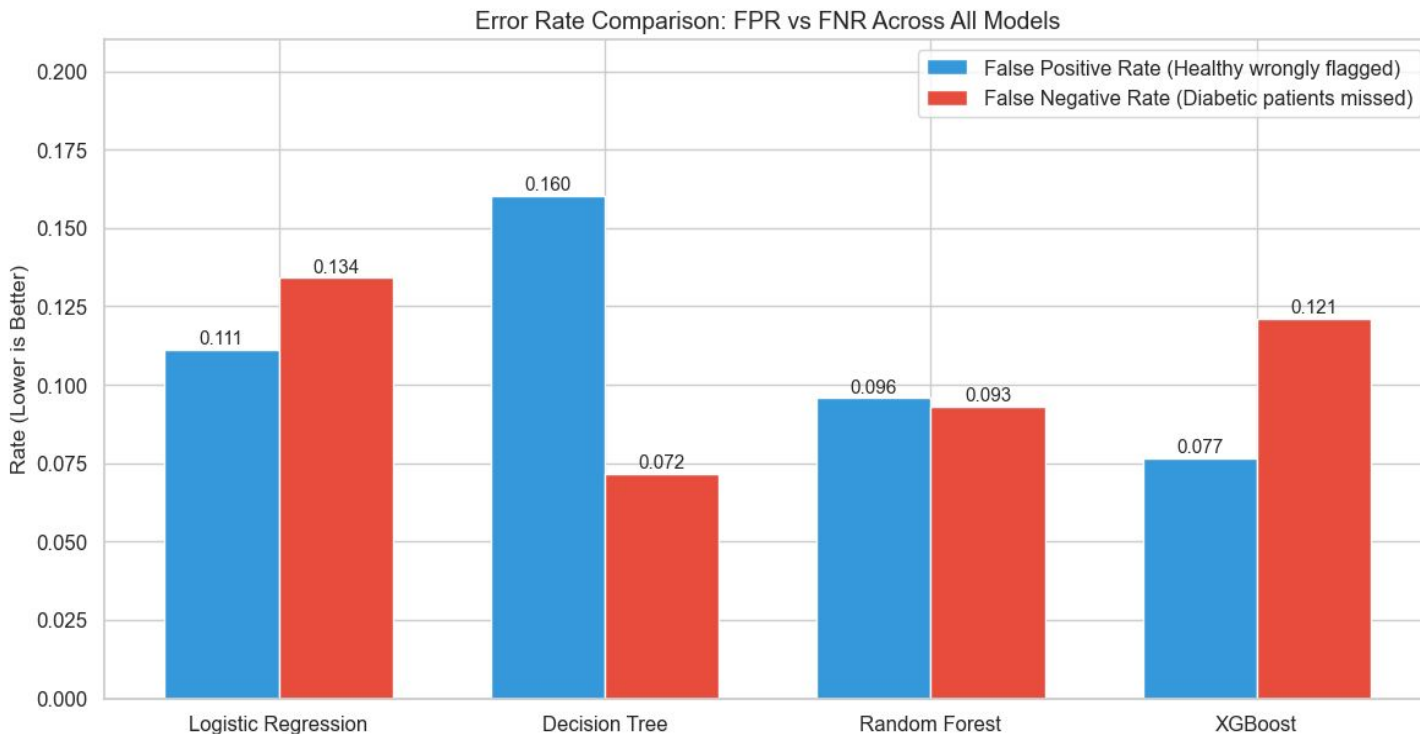


Final Choice: Class weight

Model Performance

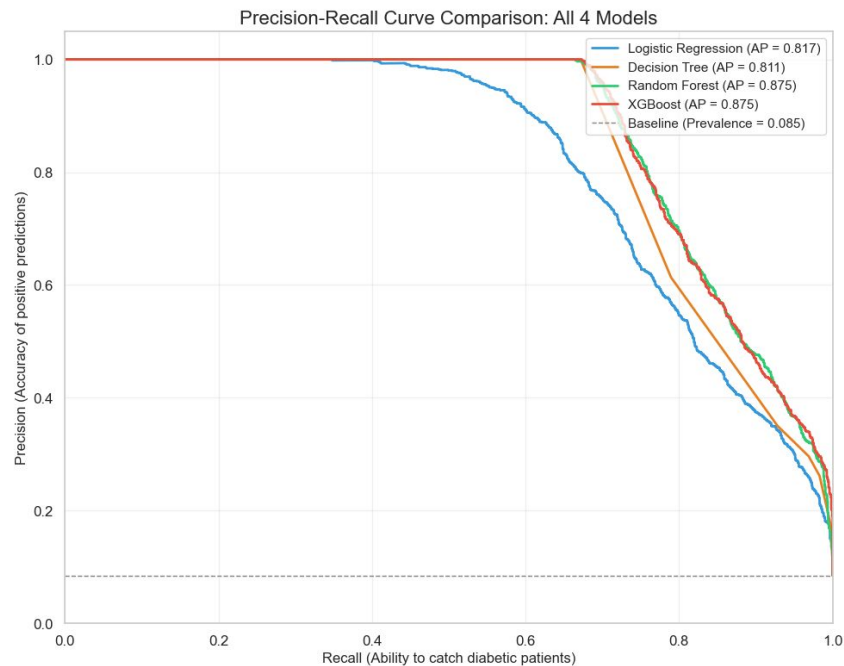
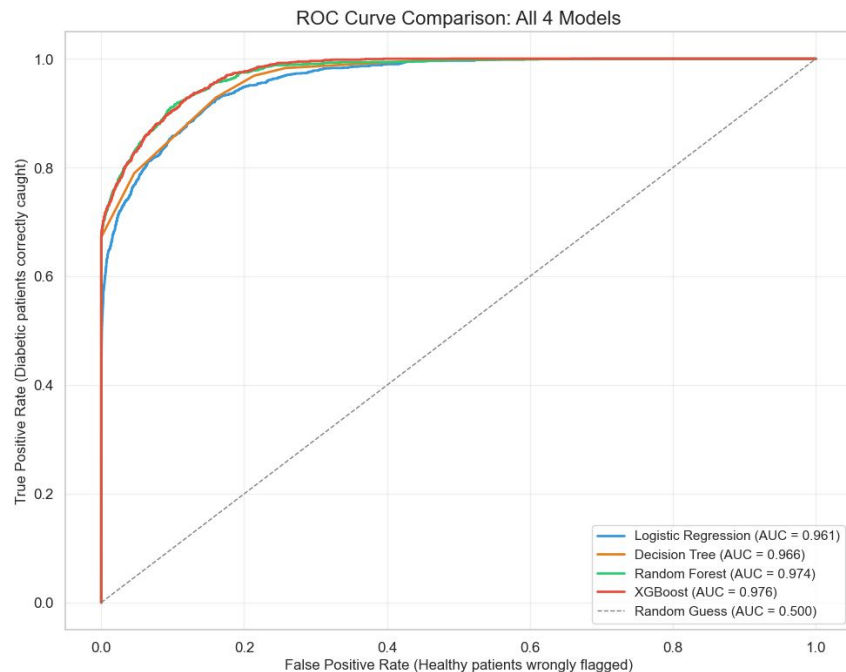


Model Performance

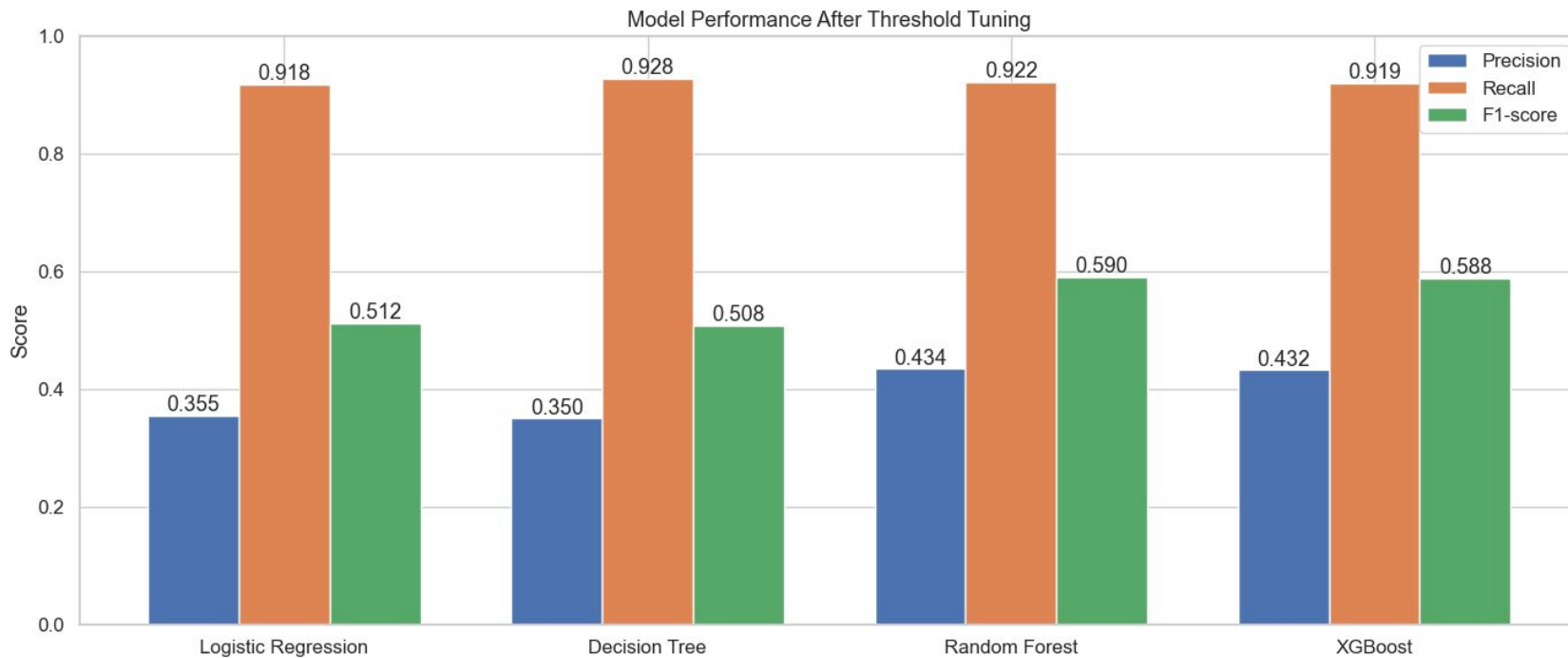


Final Choice: Decision Tree??

Model Performance

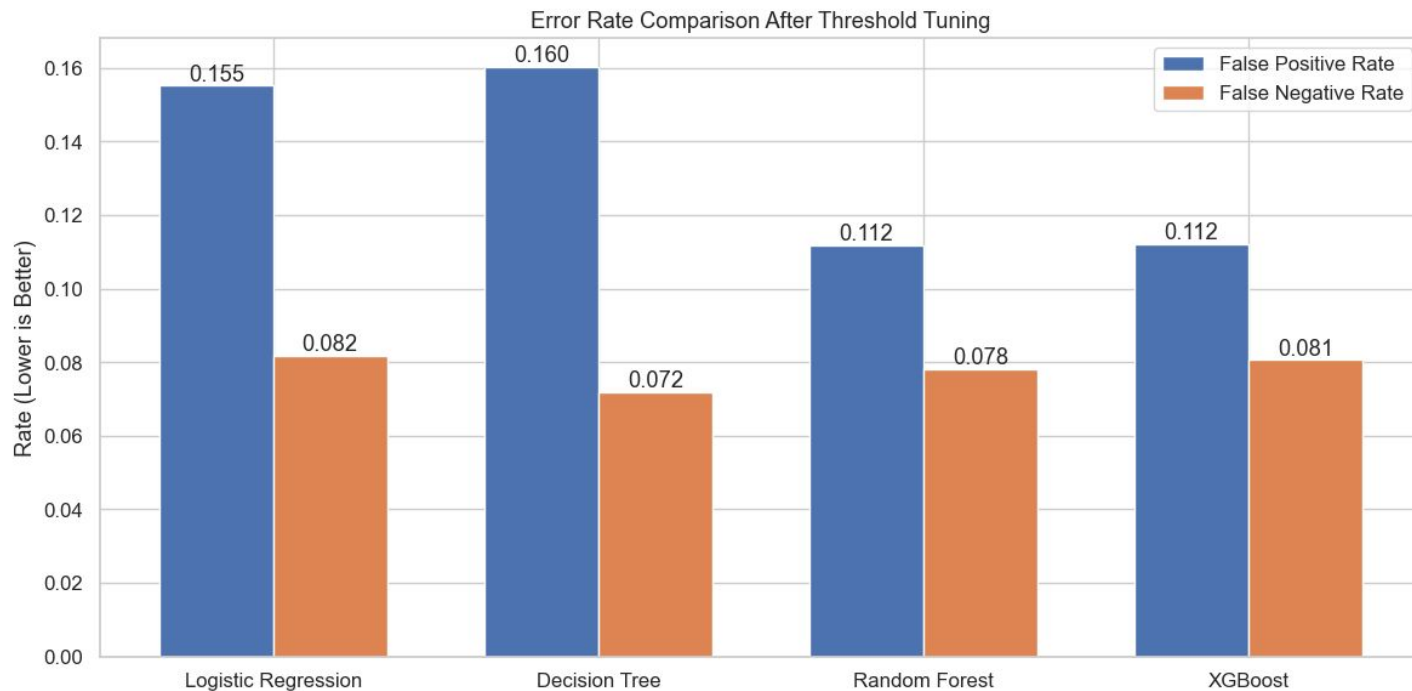


Model Comparison - Optimal Threshold



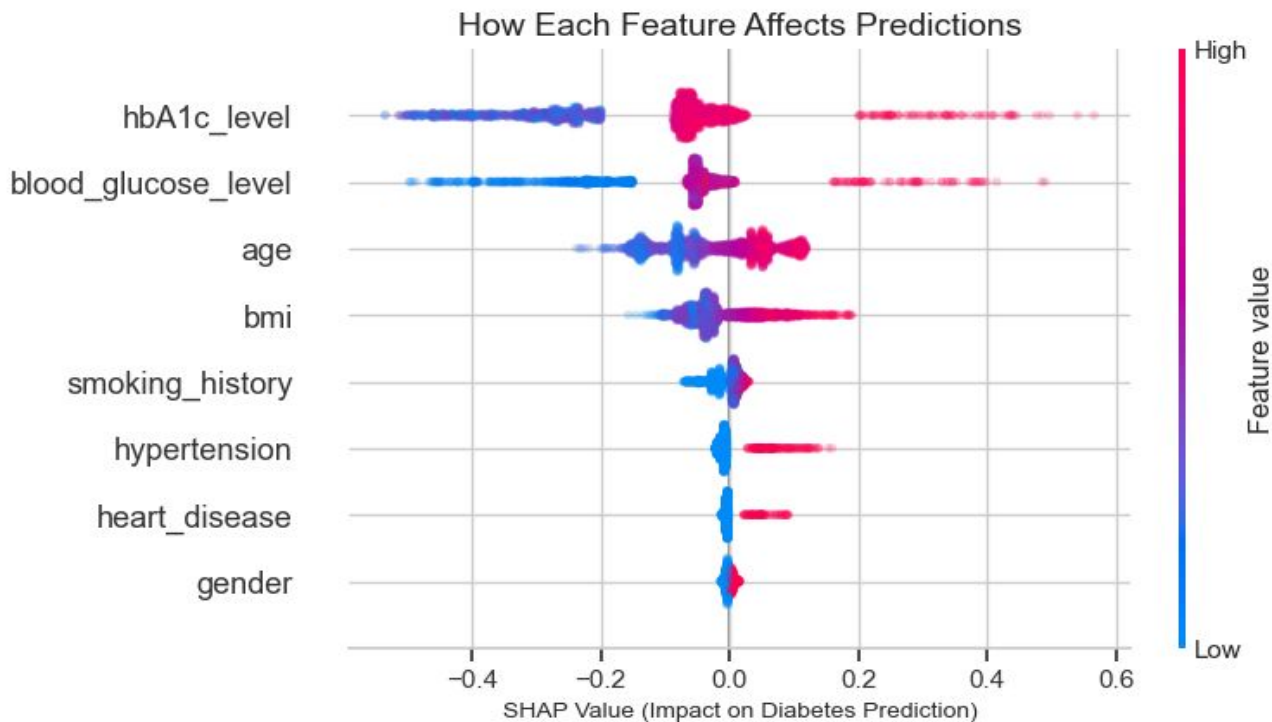
Final Choice: Random Forest

Model Comparison - Optimal Threshold



Final Choice: Random Forest

Model Diagnostics - Feature Importance



HbA1c and Glucose dominate predictions, followed by age and BMI.

Model Diagnostics - Overfitting Check

Metric	Training Set	Test Set	Difference
Accuracy	0.905	0.904	0.001
Recall	0.921	0.907	0.014

The gap between training and test performance metrics are very small, suggesting no overfitting and the model generalizes exceptionally well to unseen data.

Limitations & Future Work

Limitations

- **Clinical variables included**
HbA1c and blood glucose are clinical indicators used to diagnose diabetes. They may inflate model performance.
- **Class imbalance**
The dataset contains significantly fewer diabetes cases.
- **Limited behavioral variables**
Lifestyle factors such as diet and physical activity are not included.

Future Work

- Try models without HbA1c and blood glucose variables and compare.
- Validate models on external healthcare datasets.
- Incorporate more lifestyle and behavioral features.

Key Takeaways

- Decision Tree performed best with default threshold.
- Random Forest demonstrated the best overall performance after threshold tuning, providing strong predictive accuracy while maintaining high recall for diabetic cases.
- Clinical biomarkers (HbA1c, Glucose) are far more predictive than demographics.
- Machine learning models can effectively support early diabetes screening, helping identify high-risk individuals and enabling earlier medical intervention.

Thank you !